Calutron refining

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Abstract

This document investigates refining of pure elements from general matter through a process involving the ionization of that matter and deflection of the ions in a magnetic field to separate them by mass.

The approximate energy required to separate one kilogram of matter composed in equal parts by weight of the first 104 elements of the periodic table into the individual elements via this process is 129.37 GJ.

For meteor material that energy is 310.13GJ.

It was also calculated to be 54.358GJ for kamacite of very specific composition.

Contents

1 Ionization energy

These values had been sourced from Ptable.

atomic number	element name	molar mass (gram/mol)	1st ionization energy (kJ/mol)	specific ionization energy (MJ/kg)
1	hydrogen	1.008	1312	1301.5873015873
2	helium	4.002602	2372.3	592.689455509191
3	lithium	6.94	520.2	74.9567723342939
4	beryllium	9.0121831	899.5	99.809334765957
5	boron	10.81	800.6	74.0610545790934
6	carbon	12.011	1086.5	90.4587461493631
7	nitrogen	14.007	1402.3	100.114228599986
8	oxygen	15.999	1313.9	82.1238827426714
9	fluorine	18.998403162	1681	88.4811205271337
10	neon	20.1797	2080.7	103.108569503015
11	sodium	22.98976928	495.8	21.5661146469757
12	magnesium	24.305	737.7	30.351779469245
13	aluminium	26.9815384	577.5	21.4035238257578
14	silicon	28.085	786.5	28.0042727434574
15	phosphorus	30.973761998	1011.8	32.6663580634904
16	sulfur	32.06	999.6	31.17903930131
17	chlorine	35.45	1251.2	35.2947813822285
18	argon	39.948	1520.6	38.0644838289777
19	potassium	39.0983	418.8	10.7114631582447
20	calcium	40.078	589.8	14.7163032087429
21	scandium	44.955907	633.1	14.0826877322262
22	titanium	47.867	658.8	13.7631353542106
23	vanadium	50.9415	650.9	12.7774015292051
24	chromium	51.9961	652.9	12.556710984093
25	manganese	54.938043	717.3	13.0565262399318
26	iron	55.845	762.5	13.6538633718328
27	cobalt	58.933194	760.4	12.9027454374864
28	nickel	58.6934	737.1	12.5584818736008
29	copper	63.546	745.5	11.7316589557171
30	zinc	65.38	906.4	13.8635668400122
31	gallium	69.723	578.8	8.3014213387261
32	germanium	72.63	762	10.4915324246179
33	arsenic	74.921595	947	12.6398803976344
34	selenium	78.971	941	11.9157665472135
35	bromine	79.904	1139.9	14.2658690428514
36	krypton	83.798	1350.8	16.1197164610134
37	rubidium	85.4678	403	4.71522608514552
38	strontium	87.62	549.5	6.27139922392148
39	yttrium	88.905838	600	6.7487131722441
40	zirconium	91.224	640.1	7.01679382618609
41	niobium	92.90637	652.1	7.01889439873714
42	molybdenum	95.95	684.3	7.13183949973945
43	technetium	98	702	7.16326530612245
44	ruthenium	101.07	710.2	7.0268130998318
45	rhodium	102.90549	719.7	6.99379595782499
46	palladium	106.42	804.4	7.55872956211239
47	silver	107.8682	731	6.77678871066728
48	cadmium	112.414	867.8	7.7196790435355
49	indium	114.818	558.3	4.8624780086746
50	tin	118.71	708.6	5.96916856204195
51	antimony	121.76	834	6.84954007884363
52	tellurium	127.6	869.3	6.81269592476489
53	iodine	126.90447	1008.4	7.9461346003021
54	xenon	131.293	1170.4	8.91441280190109
55	caesium	132.90545196	375.7	2.82682158225588
56	barium	137.327	502.9	3.66206208538743
57	lanthanium	138.90547	538.1	3.87385752339343
58	cerium	140.116	534.4	3.81398270004853
59	praseodymium	140.90766	527	3.74003797948245
60	neodymium	144.242	533.1	3.69587221475021

atomic number	element name	molar mass (gram/mol)	1st ionization energy (kJ/mol)	specific ionization energy (MJ/kg)
61	promethium	145	540	3.72413793103448
62	samarium	150.36	544.5	3.62130885873903
63	europium	151.964	547.1	3.60019478297492
64	gadolinium	157.25	593.4	3.77360890302067
65	terbium	158.925354	565.8	3.56016196131927
66	dysprosium	162.5	573	3.52615384615385
67	holmium	164.930329	581	3.52269957577057
68	erbium	167.259	589.3	3.52327826903186
69	thulium	168.934219	596.7	3.53214407082321
70	ytterbium	173.045	603.4	3.48695426045248
71	lutetium	174.9668	523.5	2.99199619584973
72	hafnium	178.486	658.5	3.68936499221227
73	tantalum	180.94788	761	4.20563092532502
74	tungsten	183.84	770	4.18842471714534
75	rhenium	186.207	760	4.08147921399303
76	osmium	190.23	840	4.41570730168743
77	iridium	192.217	880	4.57815905981261
78	platinum	195.084	870	4.45961739558344
79	gold	196.96657	890.1	4.51904097228276
80	mercury	200.59	1007.1	5.02068896754574
81	thallium	204.38	589.4	2.88384382033467
82	lead	207.2	715.6	3.45366795366795
83	bismuth	208.9804	703	3.36395183471751
84	polonium	209	812.1	3.88564593301435
85	astatine	210	890	4.23809523809524
86	radon	222	1037	4.67117117117117
87	francium	223	380	1.70403587443946
88	radium	226	509.3	2.25353982300885
89	actinium	227	499	2.19823788546256
90	thorium	232.0377	587	2.52976132757737
91	protactinium	231.03588	568	2.45849259431046
92	uranium	238.02891	597.6	2.51061940333214
93	neptunium	237	604.5	2.55063291139241
94	plutonium	244	584.7	2.39631147540984
95	americium	243	578	2.37860082304527
96	curium	247	581	2.35222672064777
97	berkelium	247	601	2.4331983805668
98	californium	251	608	2.42231075697211
99	einsteinium	252	619	2.45634920634921
100	fermium	257	627	2.43968871595331
101	mendelevium	258	635	2.46124031007752
102	nobelium	259	642	2.47876447876448
103	lawrencium	266	470	1.76691729323308
104	ruthenfordium	267	580	2.17228464419476

Average specific ionization energy: 32.3424893575117 MJ/kg.

Taking a simple average of the specific ionization energy would correspond to a matter which is made of equal parts by weight of all these elements, what I refer to as "general matter".

For a more accurate result, a weighted average should be taken, with the weights corresponding to the concentration by weight of the particular element in a given mix.

2 Specific ionization energy for meteor material

Here I am taking a weighted average of the specific ionization energies presented in the previous so that the final specific energy is more relevant to asteroid mining.

atomic number	element name	normalized fraction of meteor material	adjusted specific ionization energy (MJ/kg)
1	hydrogen	0.0237970036865053	30.9738778141815
2	helium	0	0
3	lithium	1.69E-06	0.000126348716619381
4	beryllium	2.88E-08	2.87E-06
5	boron	1.59E-06	0.000117495412589677
6	carbon	0.0148731273040658	1.34540444724565
7	nitrogen	0.00138815854837948	0.138974422245487
8	oxygen	0.396616728108421	32.571705672958
9	fluorine	8.63E-05	0.00763274762371741
10	neon	0	0
11	sodium	0.0054534800114908	0.117610375152801
12	magnesium	0.118985018432526	3.6114070396081
13	aluminium	0.00902303056446659	0.193124649667102
14	silicon	0.138815854837948	3.88743705999807
15	phosphorus	0.00109069600229816	0.0356290661494892
16	sulfur	0.0396616728108422	1.23661285532495
17	chlorine	0.00036687047350029	0.0129486131577874
18	argon	0	0
19	potassium	0.000694079274189738	0.00743460457438462
20	calcium	0.0109069600229816	0.160510130783835
21	scandium	6.35E-06	8.94E-05
22	titanium	0.000535432582946369	0.00736923111214548
23	vanadium	6.05E-05	0.000772829006206731
24	chromium	0.00297462546081316	0.0373515121973554
25	manganese	0.00267716291473185	0.0349544478447686
26	iron	0.218139200459632	2.97844283911665
27	cobalt	0.000585009673959922	0.00754823090157178
28	nickel	0.0128900436635237	0.161879379698285
29	copper	0.000109069600229816	0.00127956735233261
30	zinc	0.00017847752764879	0.00247433513399913
31	gallium	7.54E-06	6.26E-05
32	germanium	2.08E-05	0.000218458656312507
33	arsenic	1.78E-06	2.26E-05
34	selenium	1.29E-05	0.000153594751077937
35	bromine	1.19E-06	1.70E-05
36	krypton	0	0
37	rubidium	3.17E-06	1.50E-05
38	strontium	8.63E-06	5.41E-05
39	yttrium	1.88E-06	1.27E-05
40	zirconium	6.54E-06	4.59E-05
41	niobium	1.88E-07	1.32E-06
42	molybdenum	1.19E-06	8.49E-06
43	technetium	0	0
44	ruthenium	8.03E-07	5.64E-06
45	rhodium	1.78E-07	1.25E-06
46	palladium	6.54E-07	4.95E-06
47	silver	1.39E-07	9.41E-07
48	cadmium	4.36E-07	3.37E-06
49	indium	4.36E-08	2.12E-07
50	tin	1.19E-06	7.10E-06
51	antimony	1.19E-07	8.15E-07
52	tellurium	2.08E-06	1.42E-05
53	iodine	2.48E-07	1.97E-06
54	xenon	0	0
55	caesium	1.39E-07	3.92E-07
56	barium	2.68E-06	9.80E-06
57	lanthanium	2.78E-07	1.08E-06
58	cerium	7.44E-07	2.84E-06
59	praseodymium	9.72E-08	3.63E-07
60	neodymium	4.96E-07	1.83E-06

atomic number	element name	normalized fraction of meteor material	adjusted specific ionization energy (MJ/kg)
61	promethium	0	0
62	samarium	1.69E-07	6.10E-07
63	europium	5.85E-08	2.11E-07
64	gadolinium	2.28E-07	8.61E-07
65	terbium	3.87E-08	1.38E-07
66	dysprosium	2.68E-07	9.44E-07
67	holmium	5.85E-08	2.06E-07
68	erbium	1.78E-07	6.29E-07
69	thulium	2.88E-08	1.02E-07
70	ytterbium	1.78E-07	6.22E-07
71	lutetium	2.88E-08	8.60E-08
72	hafnium	1.69E-07	6.22E-07
73	tantalum	1.98E-08	8.34E-08
74	tungsten	1.19E-07	4.98E-07
75	rhenium	4.86E-08	1.98E-07
76	osmium	6.54E-07	2.89E-06
77	iridium	5.35E-07	2.45E-06
78	platinum	9.72E-07	4.33E-06
79	gold	1.69E-07	7.62E-07
80	mercury	2.48E-07	1.24E-06
81	thallium	7.73E-08	2.23E-07
82	lead	1.39E-06	4.79E-06
83	bismuth	6.84E-08	2.30E-07
84	polonium	0	0
85	astatine	0	0
86	radon	0	0
87	francium	0	0
88	radium	0	0
89	actinium	0	0
90	thorium	3.87E-08	9.78E-08
91	protactinium	0	0
92	uranium	9.72E-09	2.44E-08
93	neptunium	0	0
94	plutonium	0	0
95	americium	0	0
96	curium	0	0
97	berkelium	0	0
98	californium	0	0
99	einsteinium	0	0
100	fermium	0	0
101	mendelevium	0	0
102	nobelium	0	0
103	lawrencium	0	0
104	ruthenfordium	0	0

In the table above the normalized fraction of meteor material is just the mass fraction of the given element in a typical meteor, adjusted so that the sum of all values in that column is 1.

The adjusted specific ionization energy is the ionization energy of that material multiplied by the mass fraction.

The sum of the adjusted specific ionization energies should in that case be the energy needed to ionize meteor material per kilogram. It adds up to 77.5334091946941 MJ/kg.

Playing with the libre calc sheet used to calculate this revealed that this final sum is disproportionately inflated by the almost 2.4% of hydrogen included in the meteor material. If the hydrogen were removed and everything else adjusted up to fill in the gap, it would only be a bit over 44MJ/kg.

3 Specific ionization energy for electrolytic Kamacite

This is a particular special case for a material I use a lot. See one of my other documents "Felicity tech tree: low temperature electrolysis of mixed metal oxides" for context to this.

atomic number	element name	specific ionization energy (MJ/kg)	mass in LTE batch (kg)	normalized kamacite mass fraction	adjusted ionization energy (MJ/kg)
26	iron	13.6538633718328	22	0.940433819556142	12.8405548824704
27	cobalt	12.9027454374864	0.059	0.00252207251608238	0.0325416596498918
28	nickel	12.5584818736008	1.3	0.0555710893374084	0.697888518140094
29	copper	11.7316589557171	0.011	0.000470216909778071	0.00551642442072752
30	zinc	13.8635668400122	0.018	0.000769445852364116	0.0106672640040201
31	gallium	8.3014213387261	0.00076	3.24877137664849E-05	0.000269694200307523
32	germanium	10.4915324246179	0.0021	8.97686827758135E-05	0.000941811046057686
33	arsenic	12.6398803976344	0.00018	7.69445852364116E-06	9.72570354633828E-05
34	selenium	11.9157665472135	0.0013	5.55710893374084E-05	0.000662172127318904
42	molybdenum	7.13183949973945	0.00012	5.12963901576077E-06	3.65837621520073E-05
44	ruthenium	7.0268130998318	0.000081	3.46250633563852E-06	2.43303848775154E-05
45	rhodium	6.99379595782499	0.000016	6.83951868768103E-07	4.78341981513721E-06
46	palladium	7.55872956211239	0.000066	2.82130145866843E-06	2.13254547392678E-05
47	silver	6.77678871066728	0.000014	5.9845788517209E-07	4.05562264004404E-06
48	cadmium	7.7196790435355	0.000044	1.88086763911228E-06	1.45196944973192E-05
49	indium	4.8624780086746	0.0000044	1.88086763911228E-07	9.14567753241119E-07
50	tin	5.96916856204195	0.00012	5.12963901576077E-06	3.0619679947503E-05
51	antimony	6.84954007884363	0.000012	5.12963901576077E-07	3.51356680284534E-06
52	tellurium	6.81269592476489	0.00021	8.97686827758135E-06	6.11566739318297E-05
74	tungsten	4.18842471714534	0.000012	5.12963901576077E-07	2.14851068436455E-06
75	rhenium	4.08147921399303	0.0000049	2.09460259810232E-07	8.5490769657304E-07
76	osmium	4.41570730168743	0.000066	2.82130145866843E-06	1.24580414513036E-05
77	iridium	4.57815905981261	0.000054	2.30833755709235E-06	1.0567936500108E-05
78	platinum	4.45961739558344	0.000098	4.18920519620463E-06	1.86822523666627E-05
79	gold	4.51904097228276	0.000017	7.2669886056611E-07	3.28398192540945E-06
80	mercury	5.02068896754574	0.000025	1.06867479495016E-06	5.36548375290048E-06
81	thallium	2.88384382033467	0.0000078	3.3342653602445E-07	9.61550055449706E-07
82	lead	3.45366795366795	0.00014	5.9845788517209E-06	2.06687481963874E-05
83	bismuth	3.36395183471751	0.0000069	2.94954243406244E-07	9.92211868264151E-07

The sum of adjusted ionization energy for Kamacite turns out to be 13.589417469546 MJ/kg.

4 Calutron efficiency

The total efficiency of the calutron here means the ratio of energy per kilogram that is required to ionize the material against the energy that is actually used by the caluton per kilogram of material refined.

It should be a product of all the following factors multiplied together:

- Efficiency of the ion source:

  The best method for generating ions seems to be Electron Ionization achieved by bombarding the material sample with electrons. Judging from the following sentence given on wikipedia, I will set the efficiency at 0.1%. "Under these conditions, about 1 in 1000 analyte molecules in the source are ionized."

- Efficiency of the ion accelerator: unknown, for now assumed 50%.

- Efficiency multipllier of by the deflector magnet: unknown, for now assumed 50%.

This gives a final efficiency of 0.025%.

$$\eta = 0.001 \cdot 0.5 \cdot 0.5 = 0.00025$$

Correspondingly the specific energy of refining general matter (composed in equal parts by weight of the first 104 elements of the periodic table) would be approximately 129.37 GJ/kg.

$$\epsilon = \frac{32.3424893575117 MJ/kg}{0.00025} = 129.3699574 GJ/kg$$

For meteor material the energy per kilogram to be refined this way is approximately 310.13GJ/kg.

$$\epsilon = \frac{77.5334091946941 MJ/kg}{0.00025} = 310.1336368 GJ/kg$$

For electrolytic kamacite the energy per kilogram to be refined this way is 54.358GJ/kg.

$$\epsilon = \frac{13.589417469546 MJ/kg}{0.00025} = 54.35766988 GJ/kg$$